VHDL Assignment 2 - 8-to-1 Multiplexer

Objective: Implement an 8-to-1 multiplexer in VHDL, import it into Logisim-evolution, and verify that the output follows the selected data input according to the three select signals.

1. VHDL Source Code

```
LIBRARY ieee;
USE ieee.std_logic_1164.all;

ENTITY mux8to1 IS
    PORT (
        d0 : IN std_logic;
        d1 : IN std_logic;
        d2 : IN std_logic;
        d3 : IN std_logic;
        d4 : IN std_logic;
        d5 : IN std_logic;
        d6 : IN std_logic;
        d7 : IN std_logic;

        s0 : IN std_logic;
        s1 : IN std_logic;
        s2 : IN std_logic;

        y  : OUT std_logic
    );
END mux8to1;

ARCHITECTURE Behavioral OF mux8to1 IS
    SIGNAL sel : std_logic_vector(2 DOWNTO 0);
BEGIN
    sel <= s2 & s1 & s0;

    PROCESS(d0, d1, d2, d3, d4, d5, d6, d7, sel)
    BEGIN
        CASE sel IS
            WHEN "000" => y <= d0;
            WHEN "001" => y <= d1;
            WHEN "010" => y <= d2;
            WHEN "011" => y <= d3;
            WHEN "100" => y <= d4;
            WHEN "101" => y <= d5;
            WHEN "110" => y <= d6;
            WHEN "111" => y <= d7;
            WHEN OTHERS => y <= '0';
        END CASE;
    END PROCESS;
END Behavioral;
```

2. Logisim-evolution Simulation

The data inputs were set to d0=0, d1=1, d2=0, d3=1, d4=0, d5=1, d6=0, and d7=1. The select signals s2, s1, and s0 were changed to confirm that the corresponding data input appears at the output y.

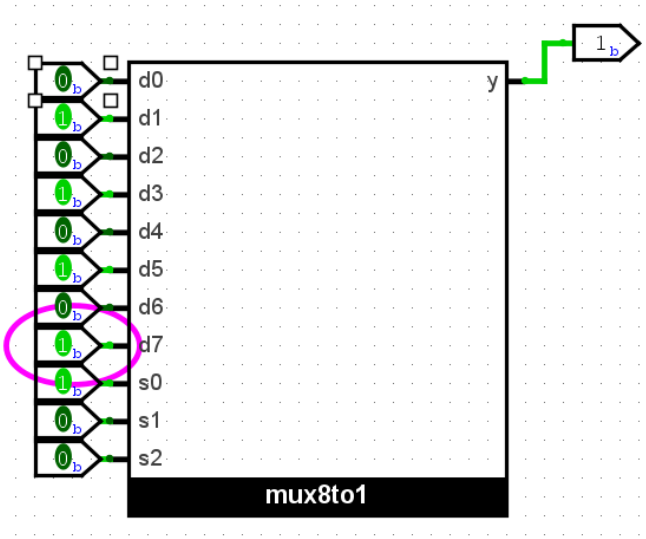


Figure 1. s2s1s0 = 001 selects d1. Since d1 = 1, the output is y = 1.

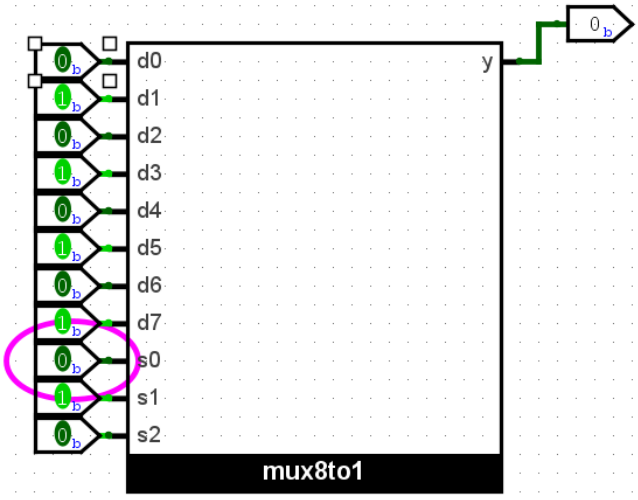


Figure 2. s2s1s0 = 010 selects d2. Since d2 = 0, the output is y = 0.

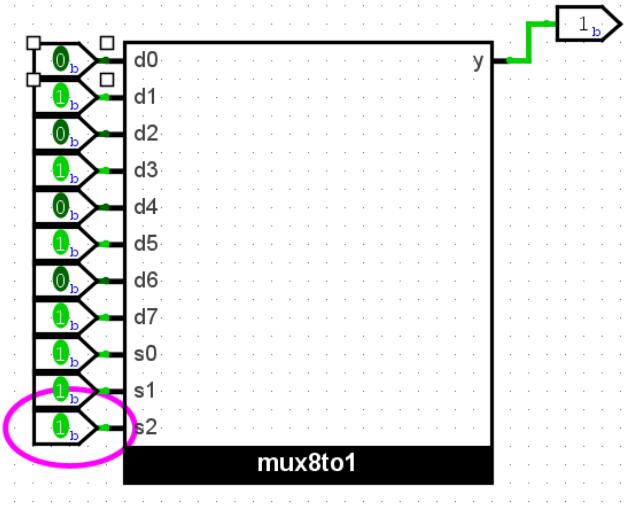


Figure 3. s2s1s0 = 111 selects d7. Since d7 = 1, the output is y = 1.

3. Verification Results

s2	s1	s0	Selected input	Input value	Output y	Result
0	0	1	d1	1	1	Pass
0	1	0	d2	0	0	Pass
1	1	1	d7	1	1	Pass

4. Conclusion

The 8-to-1 multiplexer was successfully implemented in VHDL and simulated in Logisim-evolution. For each tested select value, the output y matched the value of the selected data input. Therefore, the designed VHDL module operates correctly as an 8-to-1 multiplexer.